

SIMATS ENGINEERING

CO5- ASSESSMENT TOOL 2 – Technical Presentation

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 5

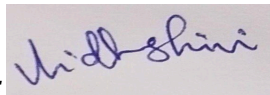
Total Marks:50

Code of Conduct:

I Vidhushini.T.S (192511061) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



1. System Analysis Presentation

Present a reverse engineering analysis of an AI-based system, explaining its architecture, components, and working process using diagrams.

2. Architecture Reconstruction Demo

Prepare a technical presentation to reconstruct the architecture of a given software/AI system, highlighting module interactions and data flow.

3. Algorithm & Logic Identification

Present how you identified algorithms and decision-making logic in a reverse-engineered system, including methodology and tools used.

4. Data Flow & Processing Pipeline

Explain through a presentation the data flow, processing pipeline, and communication mechanisms in a reverse-engineered distributed system.

5. Enhancement & Redesign Proposal

Deliver a technical presentation proposing improvements or redesign strategies for an existing system based on reverse engineering findings.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



Enhancement & Redesign Proposal

Improving the Architecture of an AI-Based Chatbot System

Course: Artificial Intelligence — CSA1709 ·

Assessment: CO5 — Technical Presentation

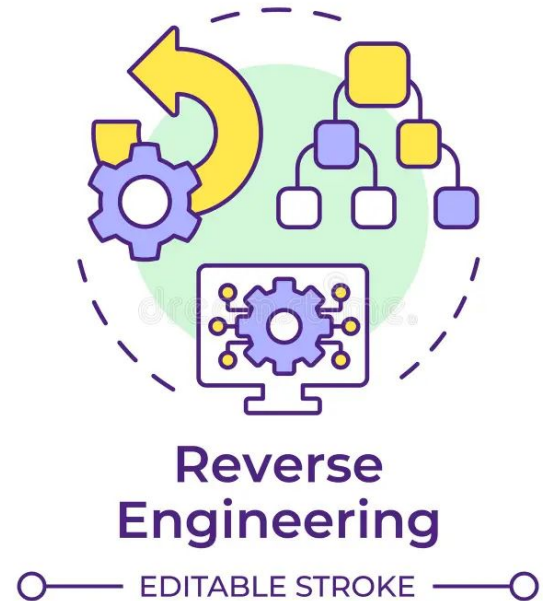
Presented by: Vidhushini.T.S

Register No: 192511061

Question:

5. Enhancement & Redesign Proposal

Deliver a technical presentation proposing improvements or redesign strategies for an existing system based on reverse engineering findings.



Existing System: Overview & Key Problems

The current chatbot pipeline: User input → Language processing (NLP) → AI/ML model inference → data access → response generation. While functional, the implementation exposes several architectural weaknesses that impact quality, performance, and operational readiness.



Response Quality

Output reliability fluctuates with the breadth and structure of available knowledge sources.



Performance & Load

High concurrent users create processing spikes; inference becomes a bottleneck without caching or load management.



Data Access

Unoptimized database calls and single-point retrieval slow responses and reduce throughput.



Security

Insufficient authentication and coarse access controls expose user data and system endpoints.



Observability

Poor logging and monitoring impede incident detection, root-cause analysis, and SLA enforcement.



Maintainability

Tight coupling between components makes targeted updates and testing risky and slow.

- ❑ Problem statement: How do we redesign the architecture to improve response quality, performance, scalability, security, and maintainability while preserving existing investments?

Proposed Enhancements — Strategic Summary

Targeted improvements map directly to identified problems. Each change is incremental, testable, and backward-compatible.

Response Quality

Introduce an organized Knowledge Retrieval layer (semantic indexing, embeddings, source ranking) to feed the model curated context.

Performance

Implement multi-tier caching (query results, model responses), batch processing, and optimized inference pipelines.

Scalability

Adopt modular microservices and autoscaling for stateless components; isolate stateful stores with sharding/replication.

Security

Add authentication (OAuth2 / mTLS), fine-grained RBAC, input sanitization, and encrypted storage for PII.

Monitoring

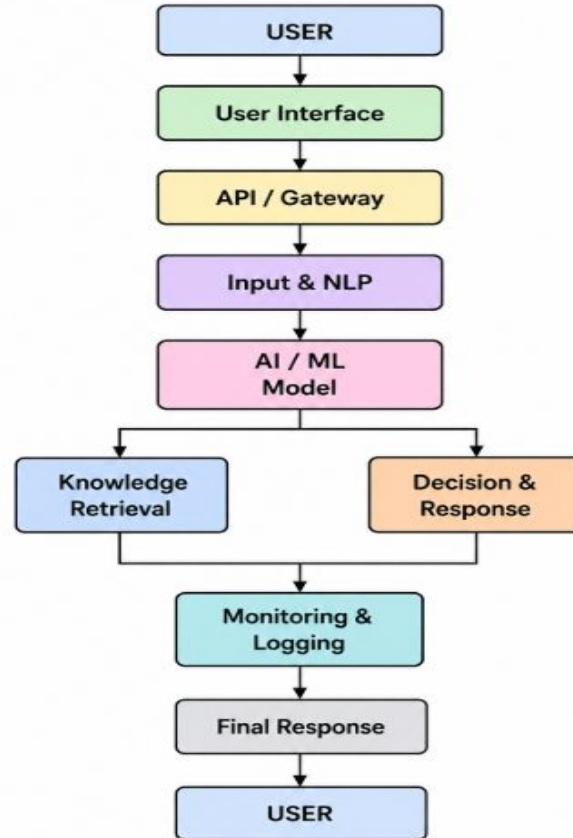
Centralized logging, distributed tracing, and real-time metrics (latency, error rates, throughput) with alerting.

Maintenance

Separate concerns into independently deployable modules with CI/CD, canary releases, and contract tests.

Workflow: Existing System → Identify Problems → Redesign Components → Improved System

Proposed Redesign Architecture



Key Improvements & Benefits

Knowledge Retrieval

Delivers relevant, ranked context to the model (embeddings + vector DB).
Expected: higher response relevance and fewer hallucinations.

Performance & Scalability

Caching reduces repeated computation; modular services scale horizontally to meet demand spikes.

Security

Authentication, RBAC, and encrypted stores protect user data and minimize exposure risk.

Monitoring & Maintenance

Structured logs, distributed tracing, and health metrics enable rapid diagnosis and safe updates via CI/CD.

❑ Expected outcome: Better responses → Faster processing → Higher scalability → Improved security → Easier maintenance

Before vs After: Feature Comparison

Comparison Table

Feature	Existing System
Architecture	Monolithic or tightly coupled components
Knowledge	Basic ad-hoc data access
Performance	Standard single-stage processing
Scalability	Limited, vertical scaling
Security	Basic protection
Monitoring	Minimal logging
Maintenance	Hard to update components independently

Proposed System

Feature	Proposed System
Architecture	Modular microservices, API Gateway, independent deploys
Knowledge	Dedicated retrieval layer (vector DB, semantic ranking)
Performance	Caching, optimized inference, batching
Scalability	Horizontal auto scaling per service
Security	OAuth2/mTLS, RBAC, encrypted PII
Monitoring	Centralized logs, tracing, SLA dashboards
Maintenance	CI/CD, canary releases, contract tests

Net result: reduced latency, higher relevance, stronger security, and simpler operations.



Conclusion & Next Steps

Reverse engineering the existing chatbot clarified core limitations. The proposed redesign introduces a Knowledge Retrieval layer, modular services, robust security, and full observability — all designed to integrate incrementally.

01

Phase 1 — Prototype

Implement Knowledge Retrieval + caching in a staging environment and run A/B quality tests.

02

Phase 2 — Harden

Add authentication, RBAC, and instrument observability; stress-test performance under realistic loads.

03

Phase 3 — Deploy

Roll out services with CI/CD, canary releases, and operational runbooks; monitor KPIs and iterate.



Recommendation: Prioritize Knowledge Retrieval and monitoring in the first sprint to maximize immediate gains in response quality and recoverability.